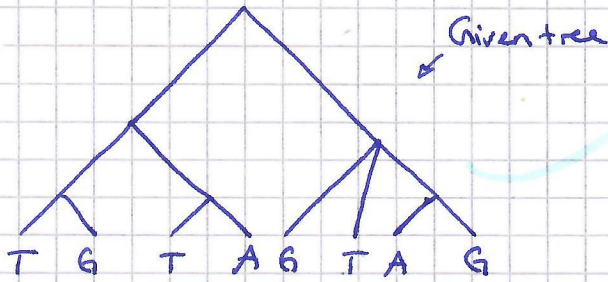


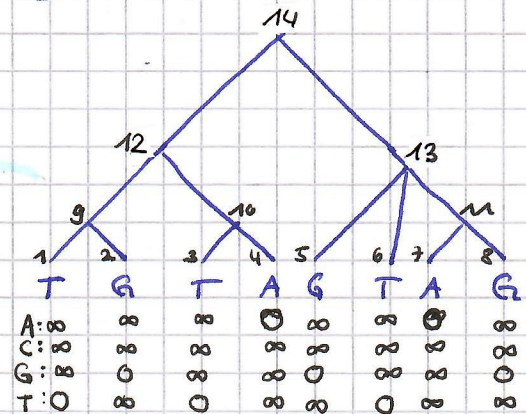
Sankoff

1. Bottom-Up

We compute the cost for the (best) labeling for each node



a) Leaves



b) Internal nodes

$$C(u, a) = \sum_{v \text{ is child of } u} \min_b (\text{cost}(a, b) + C(v, b))$$

To know which b gives the minimal cost, we compute the minimum parsimony cost for all states per node. If the node has leaves as children, we only compute for characters a child is labeled with.

$C(g, T)$ und $C(g, G)$

$$C(g, T) = \min \left\{ \begin{array}{l} (\text{cost}(T, G) + C(1, G)) \\ (\text{cost}(T, T) + C(1, T)) \end{array} \right\} \quad \text{child 1}$$

$$+ \min \left\{ \begin{array}{l} (\text{cost}(T, G) + C(2, G)) \\ (\text{cost}(T, T) + C(2, T)) \end{array} \right\} \quad \text{child 2}$$

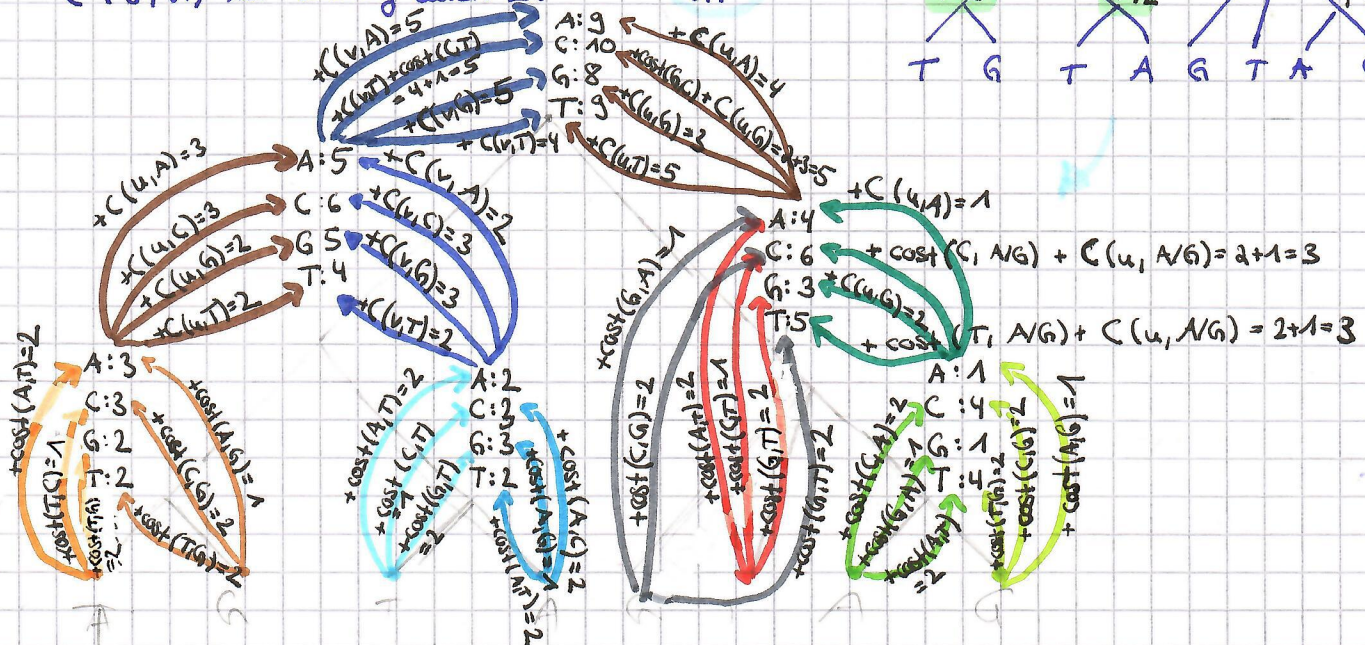
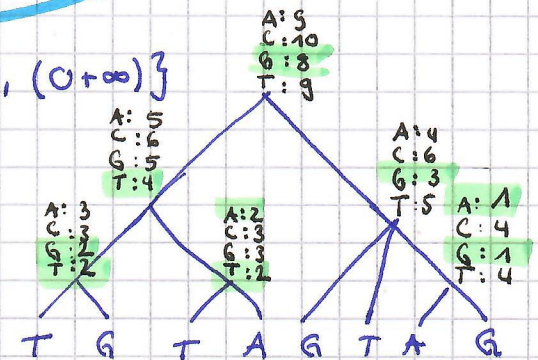
Given cost matrix:

cost	A	C	G	T
A	2	1	2	
C		2	1	
G			2	
T				2

$$= \min \{ (2 + \infty), (0 + 0) \} + \min \{ (2 + 0), (0 + \infty) \}$$

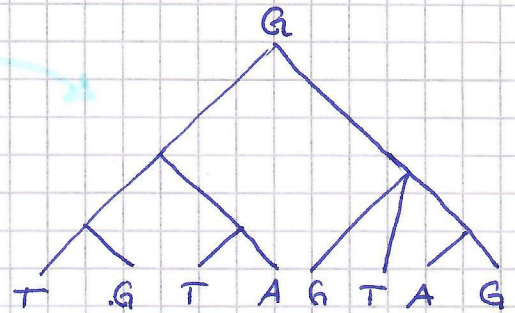
$$= (0 + 0) + (2 + 0) = \underline{\underline{2}}$$

$C(g, G)$ ist analog auch 2.



2. Top-Down Refinement

- a) Set the root state to the state with the lowest minimum parsimony cost.



- b) For all internal nodes with already labeled parent, assign the node the state that yielded the minimum in the bottom-up pass

$$s(c, \text{node}) := \text{state where } \text{cost}(\text{parent-state}, \text{state}) + C(\text{node}, \text{state})$$

$$= \min_{\text{any state}} [\text{cost}(\text{parent-state}, \text{any state}) + C(\text{node}, \text{any state})]$$

For node 13:

$$\text{cost}(G, A) + C(13, A) \stackrel{?}{=} \min \{ (\text{cost}(G, A) + C(13, A)), (\text{cost}(G, C) + C(13, C)), (\text{cost}(G, G) + C(13, G)), (\text{cost}(G, T) + C(13, T)) \}$$

$$1 + 4 \stackrel{?}{=} \min \{ (1+4), (2+6), (0+3), (2+5) \}$$

$$s \stackrel{?}{=} \min \{ 5, 8, 3, 7 \} = 3$$

Or faster: just directly calculate the minimum, here that is 3 from $\text{cost}(G, G) + C(13, G)$, therefore we set $s(c, \text{node}) = G$

Repeat for every internal node...

